

Computer Homework 8.2

Problem 1

Write and test a computer program to solve the following differential equation with initial conditions

$$\begin{cases} x' = x + e^t + tx \\ x(1) = 2 \end{cases}$$

On the interval $[1,3]$. Use the Taylor series method of order 5 and $h = 0.01$.

$$x(t+h) = x(t) + hx' + \frac{h^2}{2!}x'' + \frac{h^3}{3!}x''' + \frac{h^4}{4!}x^{(4)} + \frac{h^5}{5!}x^{(5)} + \mathcal{O}(h^6)$$

$$x'(t) = f(x, t) = x + e^t + tx$$

$$x''(t) = \frac{d}{dt}x'(t) = x' + e^t + tx' + x$$

$$x'''(t) = \frac{d}{dt}x''(t) = x'' + e^t + tx'' + x' + x' = 2x' + e^t + x'' + tx''$$

$$x^{(4)}(t) = \frac{d}{dt}x'''(t) = 2x'' + e^t + tx''' + x'' = 3x'' + e^t + x''' + tx'''$$

$$x^{(5)}(t) = \frac{d}{dt}x^{(4)}(t) = 2x''' + e^t + tx^{(4)} + x''' + x''' = 4x''' + e^t + x^{(4)} + tx^{(4)}$$

$$\therefore x_{t+1} = x_t + hx'_t + \frac{h^2}{2}x''_t + \frac{h^3}{3!}x'''_t + \frac{h^4}{4!}x^{(4)}_t + \frac{h^5}{5!}x^{(5)}_t$$

Code:

```
function Prob82n1 = Taylor5(t0,x0,b,h)
% This function uses the Taylor Series method of order 5 to find the
% solution to the differential equation x'=x + e^t + t*x, x(1) = 2.

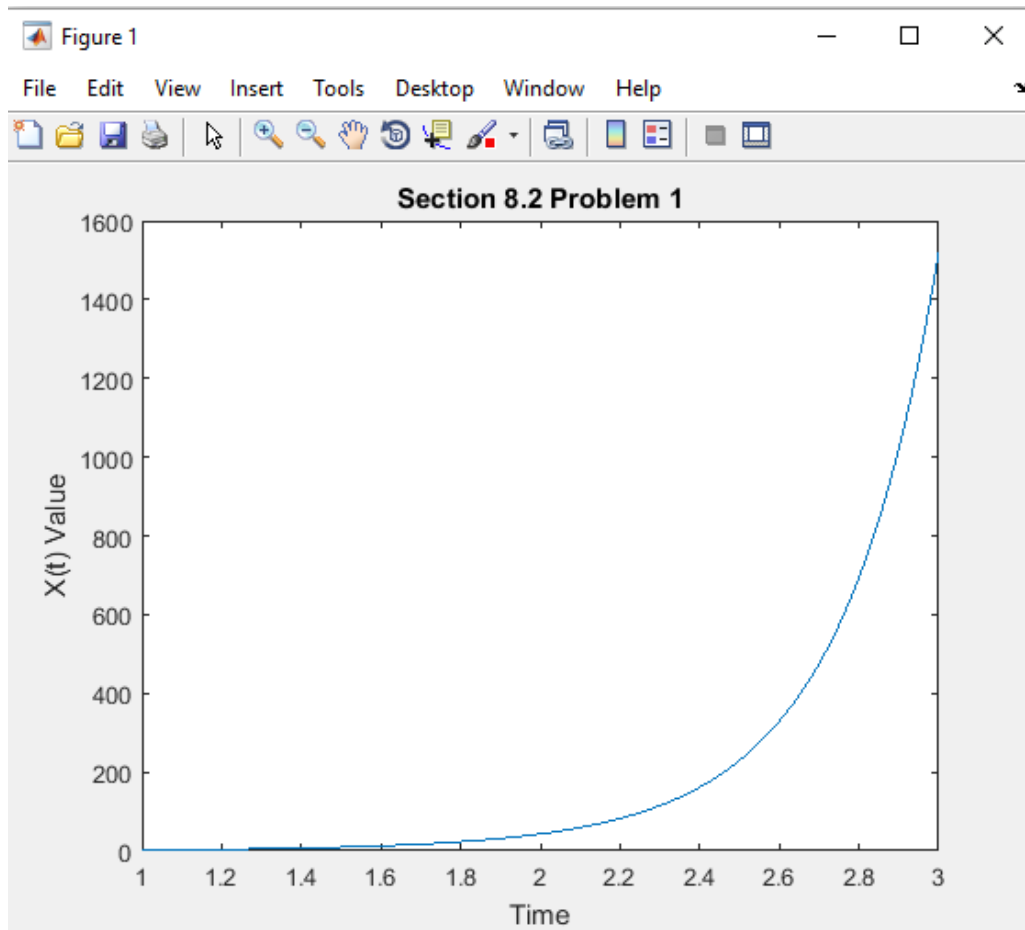
% These are the arrays that will contain all the time and x(t) values for
% the graph.
X = [x0];
T = [t0];

% These values will be used in the loop to calculate x(t+h).
t = t0;
x = x0;
xt = 0;

% This loop will go through the interval until we reach our boundary.
while (t < (b-h))
    % These are the derivatives of our function.
    x1 = x + exp(t) + t*x;
    x2 = x + x1 + exp(t) + t*x1;
    x3 = 2*x1 + x2 + exp(t) + t*x2;
```

```
x4 = 3*x2 + exp(t) + x3 + t*x3;  
x5 = 4*x2 + exp(t) + x4 + t*x4;  
  
% This is the 5th order taylor method for finding x(t+h).  
xt = x + h*x1 + x2*(h^2)/2 + x3*(h^3)/6 + x4*(h^4)/24 + x5*(h^5)/120;  
  
% Redefine the values for the next iteration.  
x = xt;  
t = t+h;  
  
% Add the values so that our array contain the solution.  
T = [T t];  
X = [X x];  
  
end  
% Now to have the plot.  
plot(T,X)  
title('Section 8.2 Problem 1')  
xlabel('Time') % x-axis label  
ylabel('X(t) Value') % y-axis label  
  
end
```

Solution:



Problem 6

The equation $\arctan\left(\frac{x}{t}\right) = \ln\sqrt{(x^2 + t^2)}$ defines x implicitly as a function of t . Verify that this implicit function is a solution to the initial-value problem

$$\begin{cases} x' = (t + x)/(t - x) \\ x(1) = 0 \end{cases}$$

$$\arctan\left(\frac{x}{t}\right) = \ln\sqrt{(x^2 + t^2)} \Rightarrow \frac{1}{1 + \frac{x^2}{t^2}} * \left(\frac{tx' - x}{t^2}\right) = \frac{1}{\sqrt{(x^2 + t^2)}} * \frac{1}{2} (x^2 + t^2)^{-\frac{1}{2}} * (2x * x' + 2t)$$

$$\Rightarrow tx' - x = x * x' + t \Rightarrow x'(t - x) = t + x$$

$$\therefore x' = \frac{t + x}{t - x}$$

$$x'' = \frac{2tx' - x}{(t - x)^2}$$

$$\arctan\left(\frac{0}{1}\right) = \ln\sqrt{0^2 + 1^2} = 0$$

To use Taylor method order 5:

$$x''' = \frac{2t(tx'' + 2x'^2 - 2x') - 2x(tx'' + 2x' - 2)}{(t - x)^3}$$

$$x'''' = \dots$$

$$x_{t+1} = x_t + hx'_t + \frac{h^2}{2}x''_t + \frac{h^3}{3!}x'''_t + \frac{h^4}{4!}x^{(4)}_t$$

[Did not finish]

Computer Homework 8.3

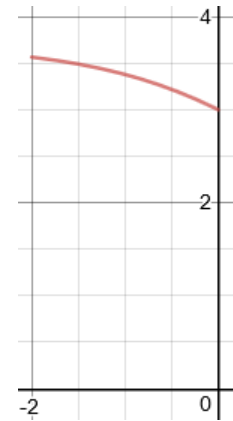
Problem 1

Write a computer problem to solve an initial-value problem $x' = f(x, t)$ with $x(t_0) = x_0$ on an interval $t_0 \leq t \leq t_m$ or $t_m \leq t \leq t_0$. Use the fourth-order Runge-Kutta method. Test it on this example:

$$\begin{cases} (e^t + 1)x' + e^t = 0 \\ x(0) = 3 \end{cases}$$

Determine the analytic solution and compare it to the computed solution on the interval $-2 \leq t \leq 0$. Use $h = 0.01$.

$$\begin{aligned} (e^t + 1)x' + e^t &= 0 \Rightarrow x' = -\frac{e^t}{e^t + 1} = -\frac{d(e^t + 1)}{e^t + 1} \\ \Rightarrow x &= -\ln(e^t + 1) + c, x(0) = -\ln(1 + 1) + c = 3 \\ \Rightarrow c &= 3 + \ln(2) \\ \therefore x &= \ln\left(\frac{2}{e^t + 1}\right) + 3 \end{aligned}$$



Code:

```
function RungeKutta4 = RK4( f,t0,x0,tm,h )
% This function uses RK4 to solve any given ODE with initial conditions.

% These are the arrays that will contain all the time and x(t) values for
% the graph.
X = [x0];
T = [t0];

% These values will be used in the loop to calculate x(t+h) or x(t-h).
t = t0;
x = x0;
xt = 0;

if (tm > t0)
    % This is for x(t+h).
    while (t < (tm-h))
        % These are the function evaluations.
        F1 = h*f(t,x);
        F2 = h*f(t + h/2,x + F1/2);
        F3 = h*f(t + h/2,x + F2/2);
        F4 = h*f(t + h,x + F3);

        % This is the RK order 4 method for finding x(t+h).
        xt = x + (F1 + 2*F2 + 2*F3 + F4)/6;
```

```
% Redefine the values for the next iteration.
x = xt;
t = t+h;
% Add the values so that our array contain the solution.
T = [T t];
X = [X x];
end
else
% This is for x(t-h) such that we work it backwards.
h = -h;
while (tm < t-h)
% These are the function evaluations.
F1 = h*f(t,x);
F2 = h*f(t + h/2,x + F1/2);
F3 = h*f(t + h/2,x + F2/2);
F4 = h*f(t + h,x + F3);

% This is the RK order 4 method for finding x(t-h).
xt = x + (F1 + 2*F2 + 2*F3 + F4)/6;

% Redefine the values for the next iteration.
x = xt;
t = t+h;

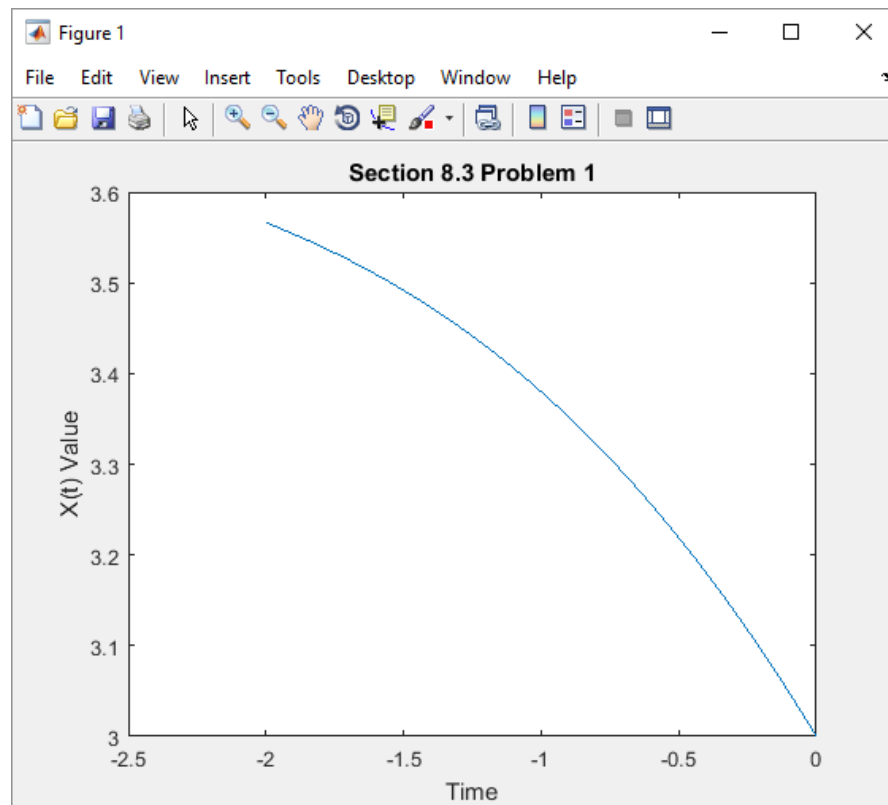
% Add the values so that our array contain the solution.
T = [T t];
X = [X x];
end
end

% Now to have the plot.
plot(T,X)
title('Section 8.3 Problem 1')
xlabel('Time') % x-axis label
ylabel('X(t) Value') % y-axis label

end
```

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Math 6313

Graph:



We can see that the true solution curve and the solution we arrived to via numerical methods is almost the same.

Problem 2

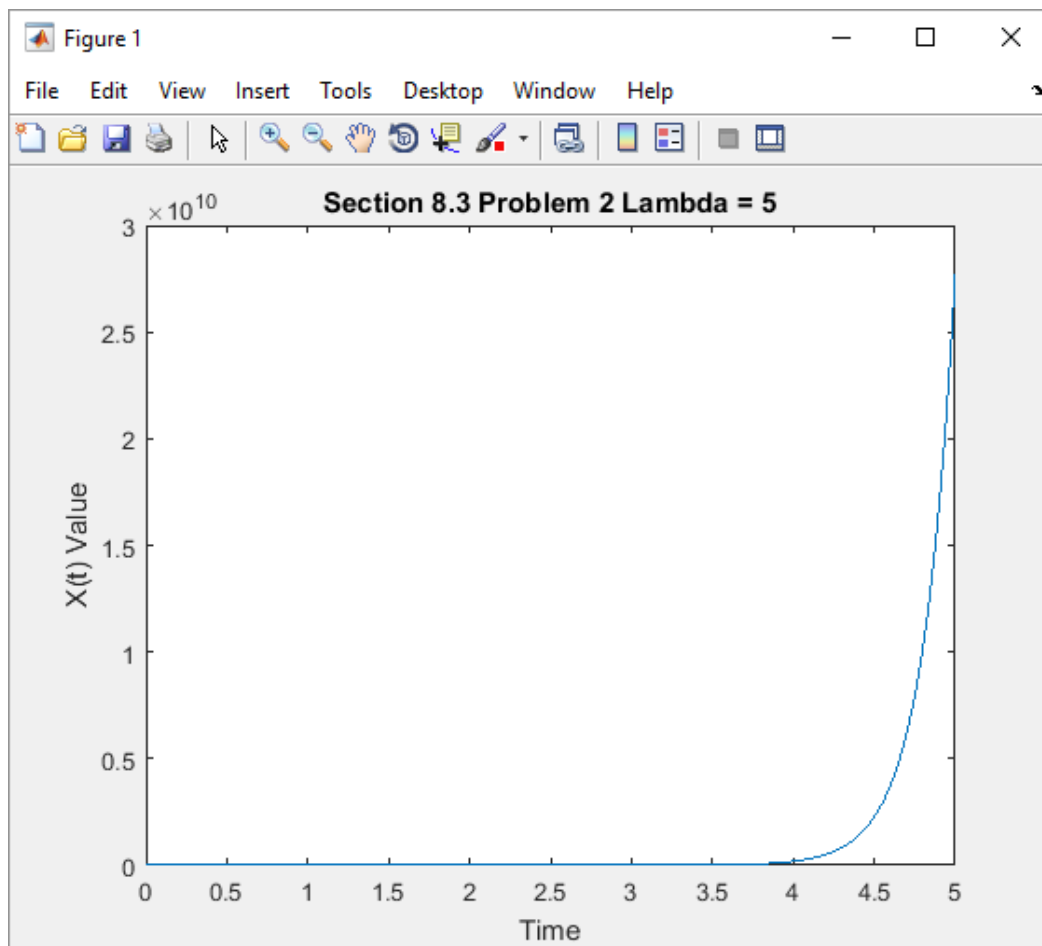
Using various values of λ , such as 5, -5 , or 10, numerically solve the following initial-value problems using the fourth-order Runge-Kutta method:

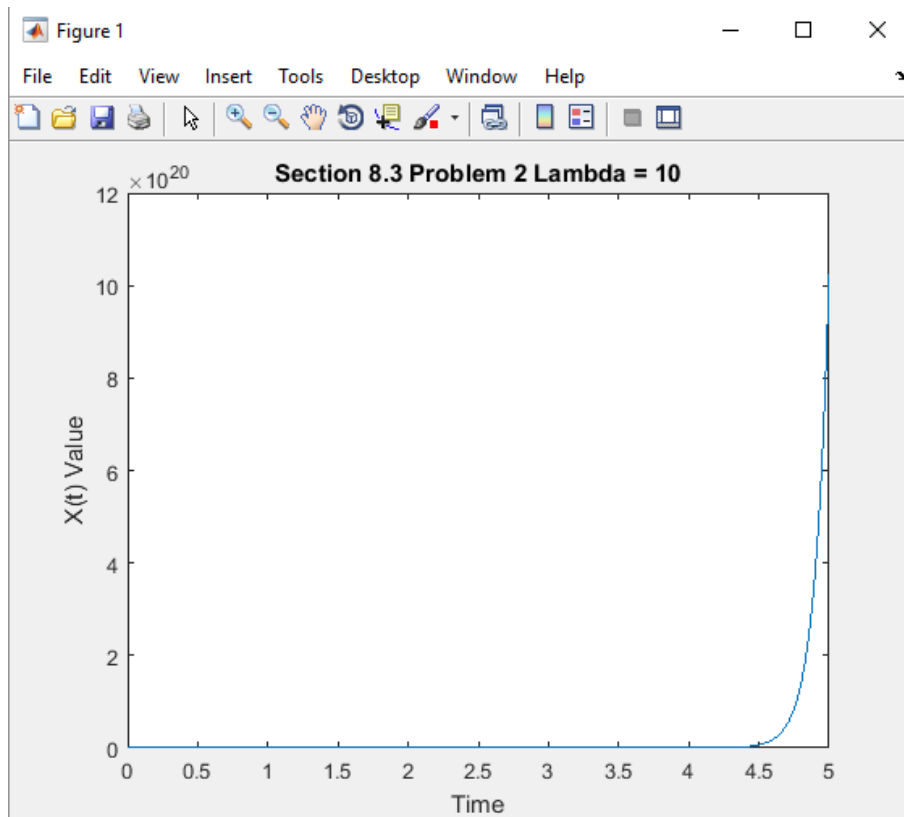
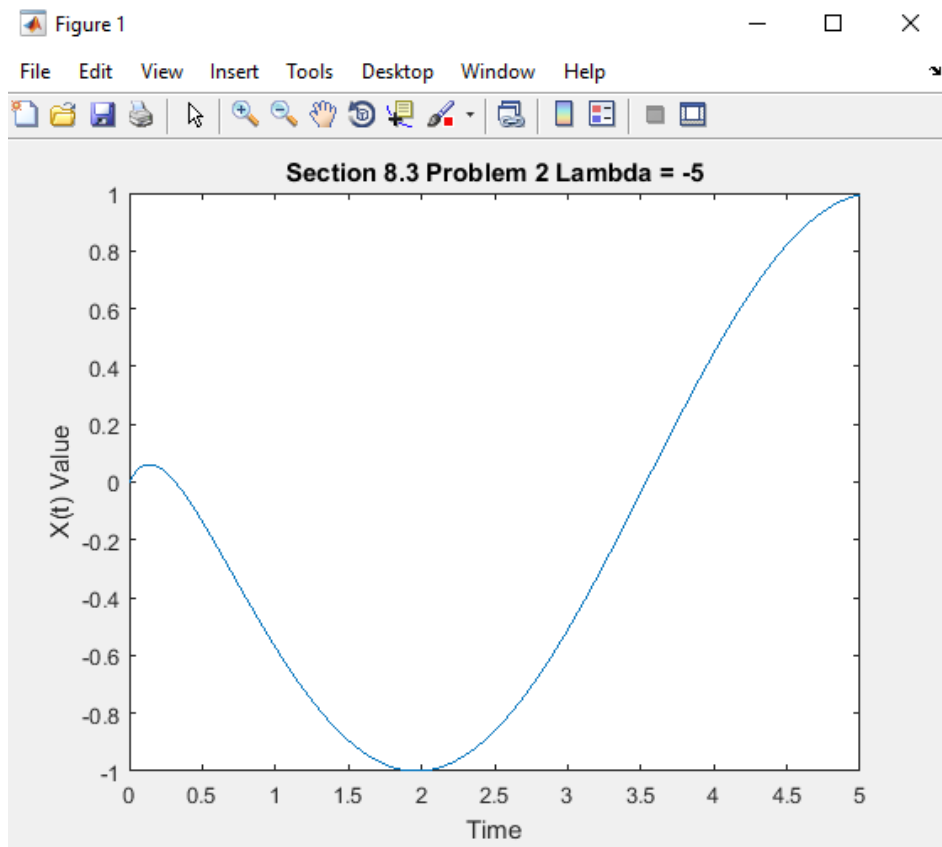
$$\begin{cases} x' = \lambda x + \cos(t) + \lambda \sin(t) \\ x(0) = 0 \end{cases}$$

Compare the numerical solution to the analytic solution on the interval $[0,5]$. Use step size $h = 0.01$. What effect does λ have on the numerical accuracy.

We use the same code as in problem 1.

Graphs:





Using wolfram to solve for the analytic solution:

$$x(t) = \frac{2\lambda e^{\lambda t} - \lambda^2 \sin(t) - 2\lambda \cos(t) + \sin(t)}{\lambda^2 + 1}$$

For smaller positive λ we see that the trigonometric functions contribute a little initially but eventually the exponential term overshadows the other terms.

Now we can see from here that as λ gets larger, we get that in the beginning, the denominator takes over thus we have that $x(t)$ is basically 0 until the exponent really starts contributing, thus we get a fast increase in the end.

For negative values of λ , we have that the exponential part does almost nothing but we have a $\lambda^2 \sin(t)$ term which gives our function a sinusoidal structure.